Scalability & Future Plan

System scaling and evolutionary roadmap

Project: AI WeatherWise API   Track: AI-ML & GEN-AI Track

Prepared: 07 October 2026   Team ID: Not provided

Maximum Marks: 1

Team Members & Assigned Roles

Team Member	Role
Jai Bharath A	Team Lead & Lead Backend/API Architect
Mukesh Balaji	Generative AI & Prompt Engineering Engineer
Yadagiri Mani	Database & Data Engineering Specialist
Deepak G	Security, Testing & Reliability Engineer
Naveen	API Integration, Documentation & Demonstration Engineer

Current Limitations

Limitation	Impact	Priority
Upstream weather rate limits	Repeated identical queries consume quota	High
Gemini free-tier quota	Burst AI calls can fail	High
Single-city query model	Route planning requires multiple calls	Medium
Single-instance deployment	Vertical limits as users grow	Medium

Scalability Plan

Area	Current	Upgrade
Weather caching	Direct upstream query	Redis cache with short TTL per city
Concurrency	Single Node.js instance	Cluster mode + horizontal autoscaling
AI caching	Generate per request	Semantic/context cache for repeated conditions
Database	MongoDB Atlas baseline	Indexes, regional partitioning/sharding when justified

Product Roadmap

Phase	Enhancement	Expected Impact
1 Current	Auth, live weather, fallback, locations, AI	Core platform
2	Redis caching + severe-weather webhooks	Lower API cost + proactive alerts
3	Route-based multi-point forecasts	Commute/logistics intelligence
4	Voice AI + Tamil/Hindi/English audio	Accessibility and hands-free use

Architectural Evolution

- Separate weatherService and aiService into independently scalable services when traffic justifies it.
- Introduce WebSockets for real-time weather/event updates.
- Add observability: structured logs, metrics, traces and alerting.
- Use rate limiting and circuit breakers around external APIs.